

VLAN Trunking and Access Ports Theory : Trunking protocols (802.1Q, ISL). Practical: Configure trunk ports on a switch. Task: Simulate communication between VLANs using trunk ports.

Q-1 In Cisco Packet Tracer, configure two switches (Switch1 and Switch2) and connect them with a crossover cable. Set the FastEthernet0/1 port on both switches as trunk ports using the 802.1Q protocol.

[A]

Objective

Connect **Switch1** and **Switch2** and configure FastEthernet0/1 on both switches as trunk ports using **802.1Q**.

Topology

Switch1 Fa0/1 ===== Fa0/1 Switch2

Trunk Link

802.1Q

Use a **crossover cable** between:

Switch1 Fa0/1

|
|

Switch2 Fa0/1

Switch1 Configuration enable
configure terminal

```
interface fastethernet0/1
switchport mode trunk no
shutdown
exit
```

end

Switch2 Configuration enable
configure terminal

```
interface fastethernet0/1
switchport mode trunk no
shutdown
```

exit

end

Verify the trunk On

both switches:
show interfaces trunk
You should see **Fa0/1** listed as a trunk port.

Result

Fa0/1 = Trunk
Encapsulation = 802.1Q
Status = Trunking

Conclusion:

The link between Switch1 and Switch2 is configured as a trunk, allowing multiple VLANs to travel through the same physical connection.

Q-2 Create three VLANs (10, 20, 30) on both switches and assign FastEthernet0/2, 0/3, and 0/4 on Switch1 to VLAN 10, 20, and 30 respectively as access ports.

[A]

Objective

Create VLANs:
VLAN 10
VLAN 20
VLAN 30

Assign the following ports on **Switch1**:

Switch1 Port	VLAN	Port Type
Fa0/2	VLAN 10	Access
Fa0/3	VLAN 20	Access
Fa0/4	VLAN 30	Access

Switch1 Configuration enable
configure terminal

vlan 10
name VLAN10

```
exit
```

```
vlan 20  
name VLAN20  
exit
```

```
vlan 30  
name VLAN30  
exit
```

```
interface fastethernet0/2 switchport  
mode access switchport access vlan  
10  
no shutdown  
exit
```

```
interface fastethernet0/3  
switchport mode access switchport  
access vlan 20  
no shutdown  
exit
```

```
interface fastethernet0/4  
switchport mode access switchport  
access vlan 30  
no shutdown  
exit
```

```
end
```

Switch2 Configuration

Create the same VLANs on Switch2:

```
enable configure terminal
```

```
vlan 10  
name VLAN10  
exit
```

```
vlan 20  
name VLAN20  
exit
```

```
vlan 30  
name VLAN30  
exit
```

```
end
```

Verify VLANs show
vlan brief You
should see:
10 VLAN10
20 VLAN20
30 VLAN30

Expected Result

Fa0/2 → VLAN 10 Fa0/3 → VLAN 20 Fa0/4 →
VLAN 30

Q-3 Connect three PCs to Switch1 on ports assigned to VLAN 10, 20, and 30. Verify that PCs in different VLANs cannot communicate with each other, but can communicate with other PCs in the same VLAN.

[A]

Connect PCs

Connect three PCs to Switch1:
PC1 → Switch1 Fa0/2 → VLAN 10
PC2 → Switch1 Fa0/3 → VLAN 20
PC3 → Switch1 Fa0/4 → VLAN 30 Assign
IP addresses.

PC1 - VLAN 10

IP Address: 192.168.10.10
Subnet Mask: 255.255.255.0

PC2 - VLAN 20

IP Address: 192.168.20.10
Subnet Mask: 255.255.255.0

PC3 - VLAN 30

IP Address: 192.168.30.10
Subnet Mask: 255.255.255.0

No default gateway is required for this test because we are testing communication **within and between VLANs without a router.**

Test 1: PC1 → PC2 On

PC1:
ping 192.168.20.10

Result

Request timed out.

**Q-4 Change the trunk encapsulation on the trunk link between Switch1 and Switch2 from 802.1Q to ISL, and test if VLAN traffic still passes between the switches.

Hint: Use the 'switchport trunk encapsulation isl' command before setting the port to trunk mode.**

[A]

Switch1:
enable
configure terminal

interface fastethernet0/1 switchport
trunk encapsulation ?

If you see:
dot1q
isl

You can configure:
interface fastethernet0/1 switchport
trunk encapsulation isl
switchport mode trunk no
shutdown
exit

Switch2:
interface fastethernet0/1 switchport
trunk encapsulation isl
switchport mode trunk no
shutdown
exit

Test the VLAN traffic

From a PC on Switch1, ping a PC belonging to the same VLAN on Switch2.

For example: PC1 → PC5 where both PCs belong to VLAN 10.

Use:

ping 192.168.1.10

If ISL is supported

The VLAN traffic should continue to pass between the switches.

If ISL is NOT supported

Packet Tracer may show an error such as an invalid/unsupported encapsulation command. In that case, **you cannot perform the ISL test using that switch model.**

Keep the trunk as 802.1Q:

```
interface fastethernet0/1 switchport
```

```
trunk encapsulation dot1q
```

```
switchport mode trunk
```

Conclusion

802.1Q is the practical choice for modern switches. ISL is an older Cisco proprietary protocol and is not supported by many modern switch models.

Q-5 Simulate a scenario where a new port on Switch2 (FastEthernet0/5) is accidentally set as a trunk port instead of an access port for VLAN 20. Observe and explain the impact on connectivity for a PC connected to that port.

[A]

Objective

Suppose Switch2 Fa0/5 was supposed to be an **access port for VLAN 20**, but someone accidentally configured it as a trunk.

Correct configuration should have been:

```
interface fastethernet0/5
```

```
switchport mode access
```

```
switchport access vlan 20
```

But accidentally we configure:

```
interface fastethernet0/5 switchport
```

```
mode trunk
```

```
no shutdown
```

Connect PC Connect:

PC → Switch2 Fa0/5

The PC is now connected to a trunk port instead of a normal access port.

What happens?

A PC normally sends **untagged Ethernet frames**. A trunk is designed primarily to carry traffic for multiple VLANs between network devices.

Therefore, configuring an end-device port as a trunk can cause unexpected connectivity behavior and VLAN assignment problems.

The PC may **not behave as expected as a normal VLAN 20 device**.

Check the configuration

Run : show interfaces trunk

You may see: Fa0/5 listed
as a trunk.

Also run: show vlan
brief

and:

show running-config interface fastethernet0/5

Correct the mistake

Change Fa0/5 back to an access port:

enable

configure terminal

interface fastethernet0/5
switchport mode access
switchport access vlan 20
no shutdown
exit

Verify:

show vlan brief

Fa0/5 should now appear under : VLAN 20

Test again

Give the PC a VLAN 20 IP address, for example:

IP Address: 192.168.20.20

Subnet Mask: 255.255.255.0 Then

ping another VLAN 20 PC:

ping 192.168.20.10

The ping should succeed if the other PC is correctly configured.